

5in1 INSULATED FUSEABLE LINKS FIRE DAMPER

El Torsion spring Fire damper



Model:
S-FSD

QCVN 06:2023/BXD

STARDUCT - Ver.05.24

Certification of inspection for fusable-link insulated fire damper [Ei] 1200x800**CÔNG BỐ KẾT QUẢ THỬ NGHIỆM****SỐ 0723A-2023/TNCL****(Ngày 27/09/2023)**

Chúng thực xác nhận, thực hiện theo

Tiêu Chuẩn Quốc Gia TCVN 9383:2012

Sản phẩm được đề cập của

CÔNG TY CỔ PHẦN ĐẦU TƯ CÔNG NGHỆ NGÔI SAO CHÂU Á

Nhà số 16, Tổ dân phố số 12 Phố Nhân Mỹ, Phường Mỹ Đình 1, Quận Nam Từ Liêm, Thành phố Hà Nội, Việt Nam

Đã được đánh giá dựa trên các yêu cầu của (các) danh mục kỹ thuật được nêu dưới đây và được chấp thuận sử dụng theo các điều kiện được đính kèm dưới đây:

Sản phẩm được chứng nhận
Mẫu van ngăn cháy hệ thống phân phối khí
mã hiệu S-FSD-L210
đạt giới hạn chịu lửa EI 123**Tiêu chuẩn áp dụng**
ISO 10294:1996**Giám đốc****Đại tá, TS. Nguyễn Thành Long**

PATENT OF FIRE DAMPER WITH FUSE LINKAGE



BỘ KHOA HỌC VÀ CÔNG NGHỆ
CỤC SỞ HỮU TRÍ TUỆ

CỘNG HÒA XÃ HỘI CHỦ NGHĨA VIỆT NAM
Độc lập – Tự do – Hạnh phúc

BẰNG ĐỘC QUYỀN

SÁNG CHẾ

Số: 40106

Tên sáng chế: VAN NGẮN CHÁY CÁCH NHIỆT CƠ CẤU CƠ KHÍ SỬ DỤNG CẦU CHỈ NHIỆT

Chủ Bằng độc quyền: Công ty Cổ phần Đầu tư Công nghệ Ngôi Sao Châu Á (VN)
Tầng 3, tòa nhà TTTM Interserco, số 17 đường Phạm Hùng, phường Mỹ Đình 1, quận Nam Từ Liêm, thành phố Hà Nội

Tác giả: Đào Huy Khánh (VN)

Số đơn: 1-2022-03810

Ngày nộp đơn: 17/06/2022

Số điểm yêu cầu bảo hộ: 06 **Số trang mô tả:** 16

Cấp theo Quyết định số: 54662/QĐ-SHTT-IT, ngày: 13/05/2024

Có hiệu lực từ ngày cấp đến hết 20 năm tính từ ngày nộp đơn (Hiệu lực bảo hộ cần duy trì hàng năm).



KT. CỤC TRƯỞNG
PHÓ CỤC TRƯỞNG



Lê Huy Anh



VN 0106

INSULATED FIRE DAMPER - Fusable link [EI 120]

S-FSD: 500(W) x 500 (H) x (210L) mm

Test Report Number: 0289-2024/TNCL



Torque fire damper vs fuse

DESCRIPTION

Starduct fire damper S-FSD is a fire fighting equipment for preventing fire, smoke and temperature spreading in HVAC systems that can remain in operation during fires.

S-FSD dampers has been matched requirements of test as regulated in the National Technical Regulation to Fire Fighting and Prevention Equipments (QCVN 032021/BCA) and have fire resistance rating of 120 minutes (Ei120) with the certificate of representative sample inspection as per the Decree 136/2020/NĐ-CP,

In addition to fire fighting and protection regulations, S-FSD dampers are also produced according to requirements of HVAC industry standards such as Smacna, AMCA...

Type: S-FSD models are symmetric dampers for installation in walls in fire zones and have fire resistance rating of 120/90/60/45/30 minutes (EI).

Ordering code:

S-FSD-L210 - WxH - TDC - EI

Fire damper code

Dimension: Width(W) x High (H)

EI Level: 30/45/60/90/120

Flange type: TDC

Product Construction

Material

- Frame: L210mm, 1 GL steel, 1.15mm + 2 layers MgSO₄ thickness 10mm, Density 950kg/m³
- Damper blade: 1 GL steel, t=1.5mm + 2 layers MgSO₄, t=10mm, density 950kg/m³ + 2 layers ceramic wool, t= 5mm, density 220kg/m³.
- Blade stop: GL steel, t= 1.15mm
- Shaft: Steel CT45 CNC, 9.5X9.5
- Bearing: Brass
- Frame sealant : Hilti Cp606

Dimension

- Max Modul size = 500(W)x500(H)x210(L)[mm]

Construction

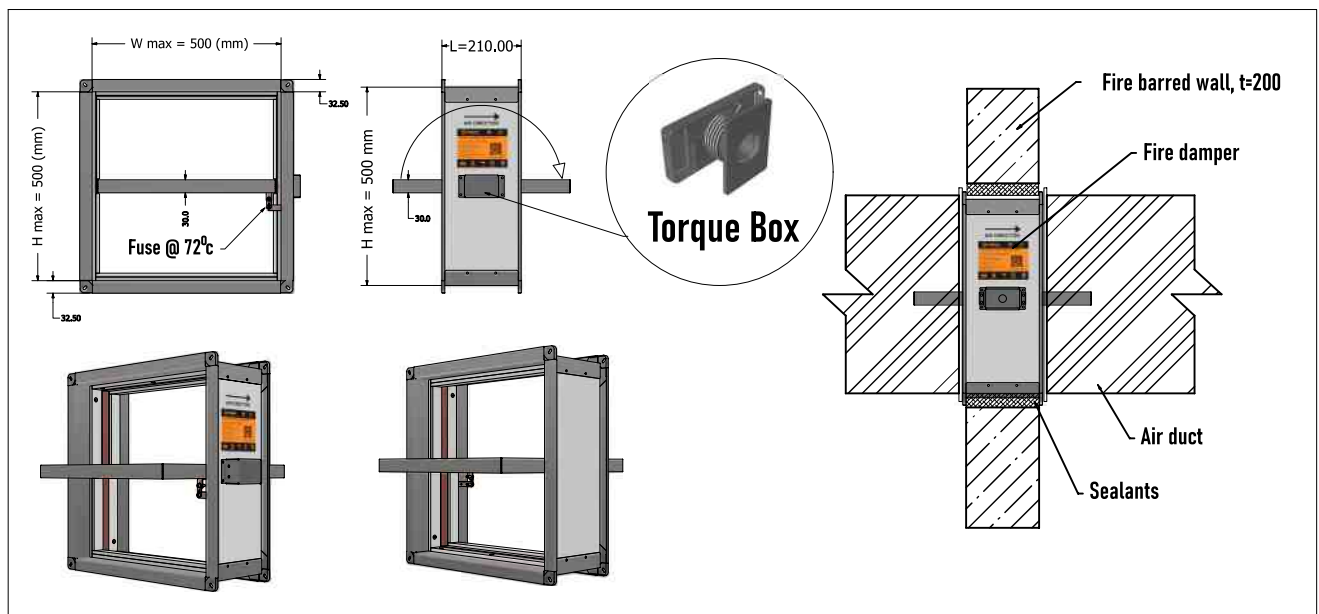
- TDC Flange
- Blade: 1 blade.
- EI level**
- EI 120 (2.0 hours).

Fuselink mechanism

- Fuse tube with torsion spring handle controller box.

Installation method : on the wall, inside the fire zone.

Construction details



INSULATED FIRE DAMPER - Fusible link [EI 120]

S-FSD: 1200(W) x 800 (H) x (210L) mm

Test Report Number: 0723-2023/TNCL



Material certificates download

DESCRIPTION

Starduct fire damper S-FSD is a fire fighting equipment for preventing fire, smoke and temperature spreading in HVAC systems that can remain in operation during fires.

S-FSD dampers has been matched requirements of test as regulated in the National Technical Regulation to Fire Fighting and Prevention Equipments (QCVN 032021/BCA) and have fire resistance rating of 120 minutes (EI120) with the certificate of representative sample inspection as per the Decree 136/2020/NĐ-CP,

In addition to fire fighting and protection regulations, S-FSD dampers are also produced according to requirements of HVAC industry standards such as Smacna, AMCA...

Type: S-FSD models are symmetric dampers for installation in walls in fire zones and have fire resistance rating of 120/90/60/45/30 minutes (EI).

Ordering code:

Product Construction

Materials

- Body: galvanized steel sheet Z18, thickness: 1.15 mm
- Insulation for body: 2 layers of MgSO₄ board (950 kg/m³), thickness 10mm and 1 layer of MgO (380kg/m³), thickness 38mm
- Blade stopper: steel sheet, thickness 1.2mm
- Axle : steel Ct45, hexagon bar, CNC lathe, D20
- Bushing: cooper
- Sealing : Hilti Cp606 glue
- Blade seal: Hilti FS one Max intumescent firestop sealant

Dimensions

- Max. Dim. module = 1200 (W) x 800 (H) x 210 (L) [mm]

Construction

- Frame bars are jointed with M8 bolts. 2-side TDC flange
- Balde: Steel frame with 2 layers of Ceramic wool, t=20mm board and 2 layers of MgSO₄ board, t=10mm.

Class

- EI 120 (2 hrs)

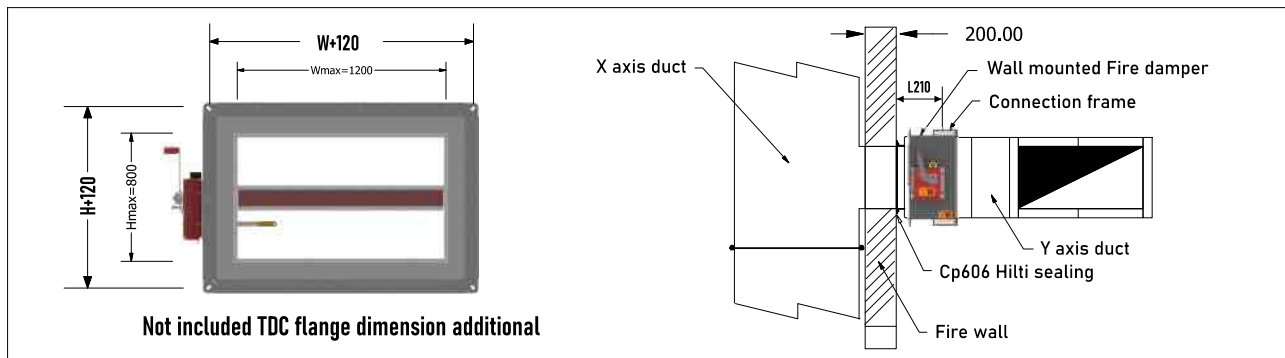
Fuselink mechanism

- Fuse tube with torsion spring handle controller box.

Installation method : on the wall, inside or outside the fire zone.

S-FSD-L210	-	WxH	-	TDC	-	EI
Fire damper code						EI Level: 30/45/60/90/120
Dimension: Width(W) x High (H)				Flange type: TDC		

Modul details



Not included TDC flange dimension additional

- Min dimension step: 50mm
- Damper max and min size (mm):
 $200 \times 200 \leq W \times H \leq 1200 \times 800$
- Overall dimension(mm) $[W+120] \times [H+120]$
- Acctuator dimension (mm) $a \times b$
- Exceeding part of the blade (x)
 - H=600 → x=200 ▪ H=650 → x=225 ▪ H=700 → x=250
 - H=750 → x=275 ▪ H=800 → x=300

Fuse tube with torsion spring contrllr 5in1

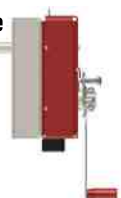
Emergency shut-off



VCD handle

**Fuse 70+/-2
or 280 C degree**

**Mechnism status
indicator**

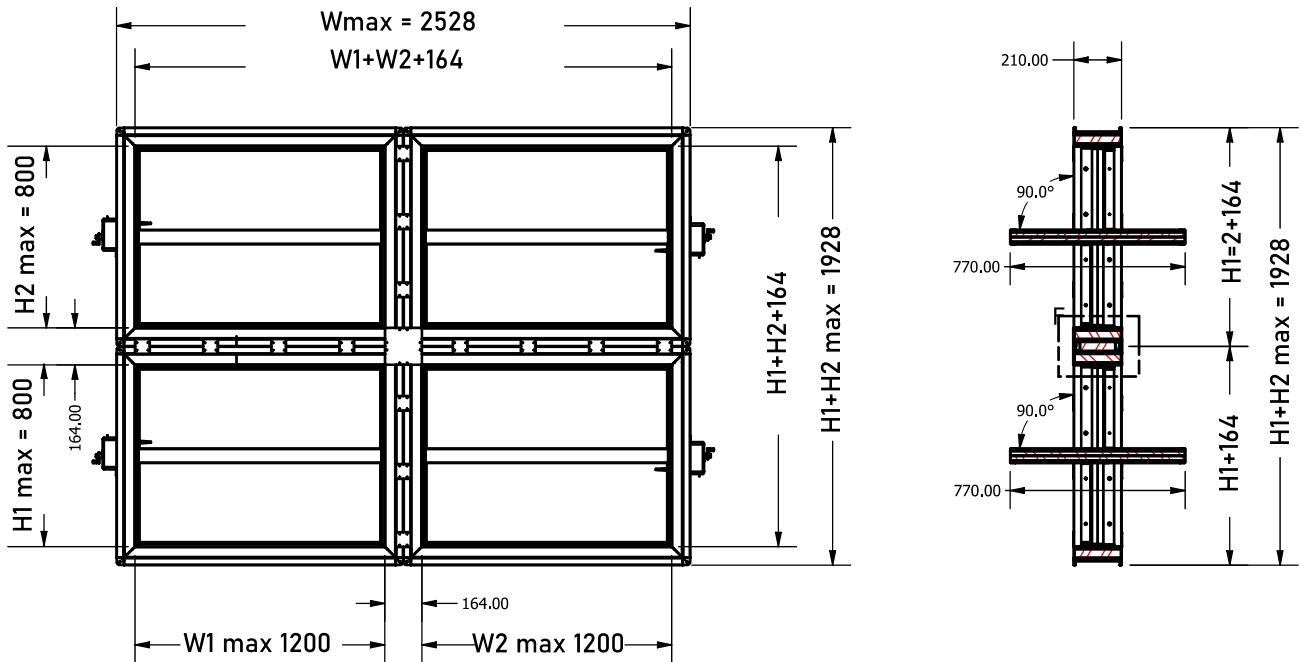


SIZE EXTENDED COMBINATION-L210

[EI]-S-FSD-L210 (Width W No limited x High H max 1764mm)

● **STANDARD BLOCK Ei120**

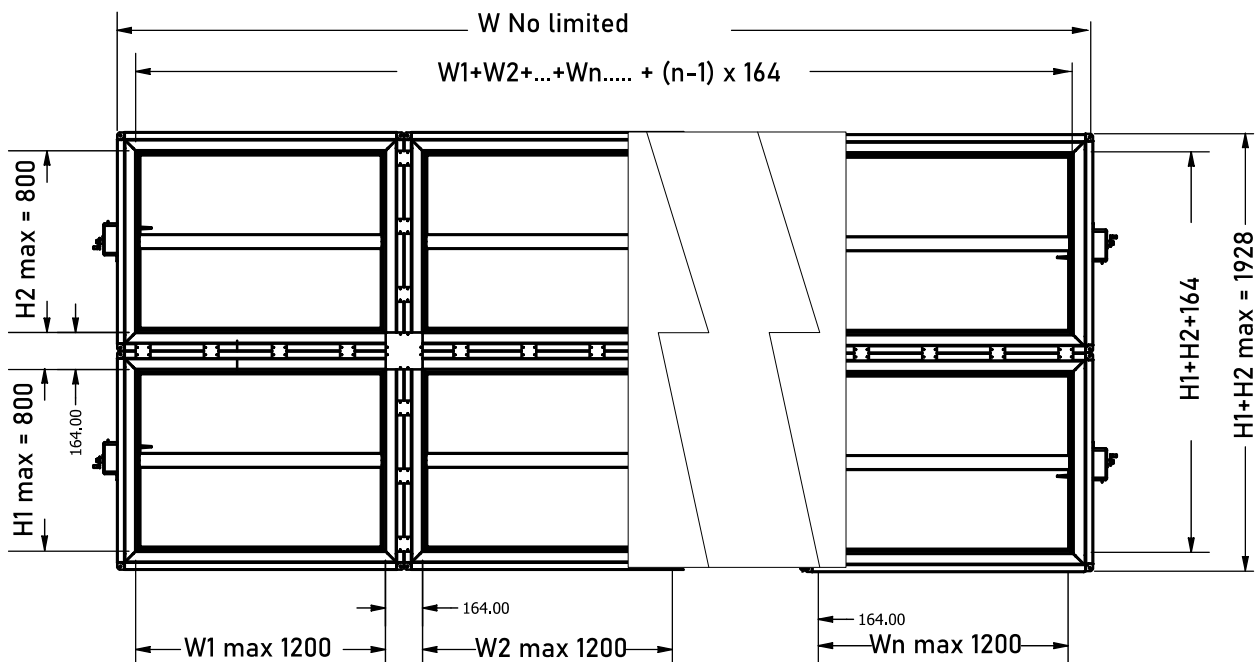
- Test Report Number : 0175-2024/TNCL



Note: The blades close clockwise from the front view of the controller on each side

● **EXPANDABLE BLOCK Ei120**

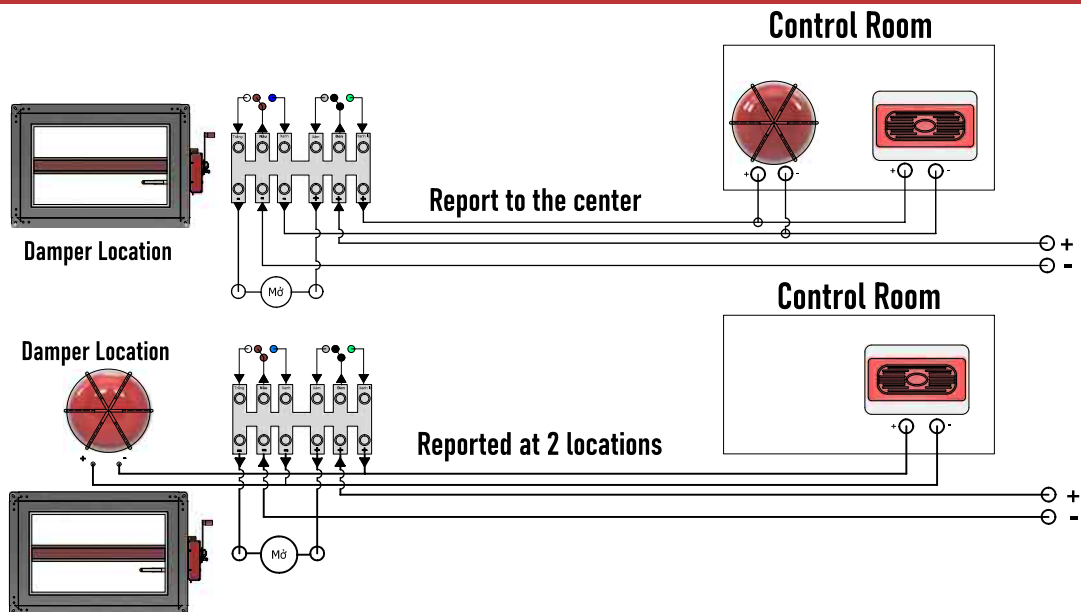
- Test Report Number : 0175-2024/TNCL



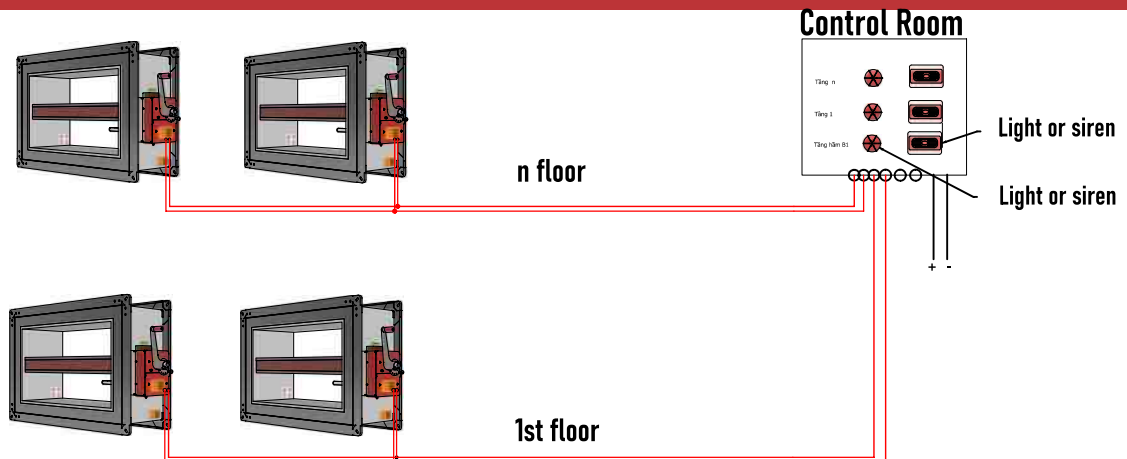
Note: Assembled from standard block at the site - Note to add motor compartment to the assemblies center (200mm for each combination)

Note: Extended functionality of 5-in-1 controller. Not required

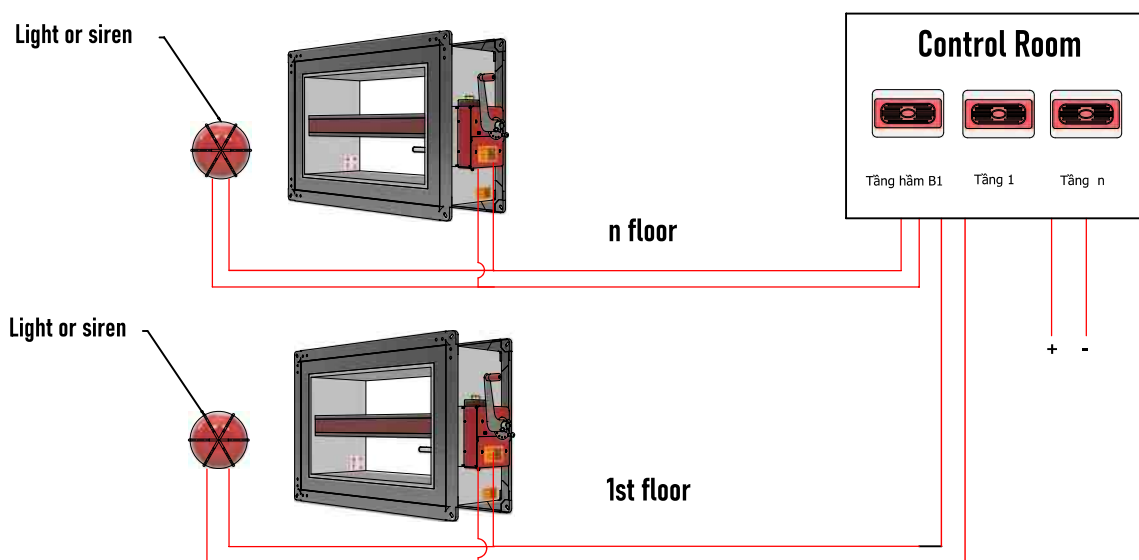
FD 5 IN 1 VALVE STATUS CONTROL WIRING DIAGRAM



Control model at the fire alarm center



Control model at the fire alarm center and damper location



MARKS AND LABELS - Applied on each damper

Nhà sản xuất : CÔNG TY CPĐTCN NGÔI SAO CHÂU Á Nơi sản xuất : Nhà máy cơ khí Starduct Cụm CN Phùng, Đan Phượng, Hà Nội, Việt Nam Hotline : 0240 3514 7999	➔	Manufacturer, made place, address, phone number.
Tên sản phẩm : VAN NGÂN CHÁY KHÔNG CÁCH NHIỆT Mã hiệu : SFD-M Cấp ngăn cháy : E120 Kích thước : W600 x H300 x L210	➔	Product name, model, dimensions, fire rate, insulation rate
Số GCN : 1021. Ngày cấp : 13.03.2023 Đơn vị cấp : Cục CS PCCC và CHCN, BCA	➔	Number of the applicable certificate, issued date, issued by
Tên dự án : BỆNH VIỆN VIỆT NHẬT Địa chỉ : Số 10 đường ABC, Phường, Quận, Huyện, Tỉnh, Thành phố	➔	Name of project, delivery address
Số quản lý : NS-S20-3-1 Ngày sản xuất : 20.03.2023	➔	Admin. ref # (Series), Prod. date
	➔	QR Code for checking origin, serial number
 Web site : www.starduct.vn	➔	Manufacturer Logo, Web address

LIST OF LEGAL DOCUMENTS ACCOMPANYING WITH GOODS FOR A PROJECT

- 1- A copy (notarized) of the certificate of verified sample
- 2- A copy (notarized) of the certificate of verified requirements for production and service
- 3- List of certified models (i.e quantity, sizes and ref. numbers of each damper in the list)
- 4- Installation drawings as per tests.
- 5- Delivery minutes from manufacturer

Warning sign system of Star Asia

 Do Not Drop Không kéo lê, tránh rơi đổ	 Do Not Walk or Stand On Không dẫm hoặc đứng lên trên	 This Side Up Xếp hướng lên trên
 Dry Storing Bảo quản nơi khô ráo	 Store In Warehouse Bảo quản trong kho	 Sharp Edge Use Glove Vật có cạnh sắc Đeo găng an toàn khi làm việc
 Handle With Care Nhẹ tay khi dịch chuyển	 Use Forklift To Move Chuyển bằng xe nâng	 Use Scissor Forklift To Lift Up Nâng lên cao bằng xe nâng chuyên dụng

S-FSD - WxH - TDC - EI

Insulated fire damper

Dim. : Width (W) x Height (H)

Flange: **TDC**

EI class: 30/45/60/90/120

[illegible]

Storage and handling

As fire damper is a safety element, it should be stored and handled with care

Avoid:

- Any kind of impact or damage
- Contact with water
- Deformation of the casing

Recommendations for Star Asia :

- To unload in a dry area
- Not to flip or roll the product to move it
- Not use the damper as a scaffold, working table etc.
- Do not store smaller damper inside large one
- Do not stand or seat on the product



Nhận và bảo quản nơi
khô ráo nước



Sử dụng gắng bảo hộ
khi làm việc



Không dẫm, đạp,
trèo, đi trên van

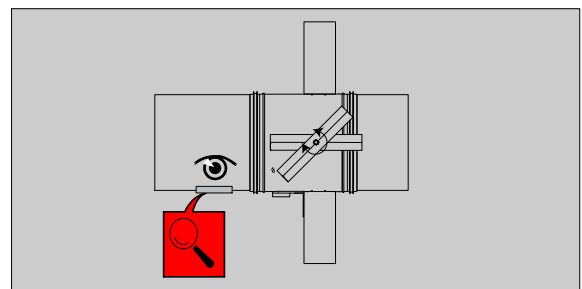
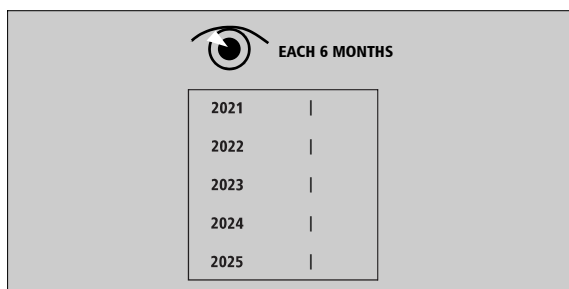


Không thả, lăn, lật
lôi, kéo, đẩy, ...

Important notes when installation

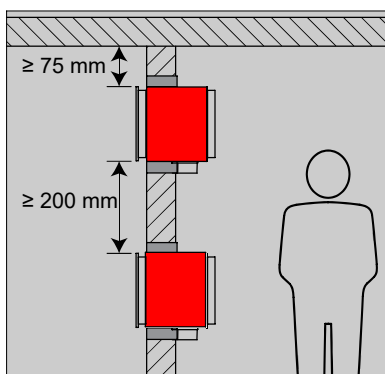
General :

- The installation must comply with the installation manual delivered with the product and the classification report.
- Axis orientation: see the declaration of performance
- Avoid obstruction of adjoining ducts.
- Product installation: always doing with closed damper blade
- Verify if the blade can move freely
- Please observe safety distances with respect to other construction elements. The operating mechanism must also remain accessible: allow for a clearance of 200 mm around the housing.
- The air tightness class will be maintained if the damper is installed according to the installation manual.
- The fire dampers are always tested in standardised constructions according to QCVN 60:2021/BXD. The achieved results are valid for similar supporting constructions with a fire resistance, thickness and density equal or superior to the supporting construction used during the test.
- The damper must remain accessible for inspection and maintenance.
- Schedule at least two running checks each year.



Installation at a minimal distance from another damper or from an adjacent supporting construction

1



1. Principle according to the test standard:

Minimum distance of ≥ 75 mm from the ceiling

Minimum distance ≥ 200 mm to an adjacent damper



Use scissor lift to lift up



Use forklift to move

